

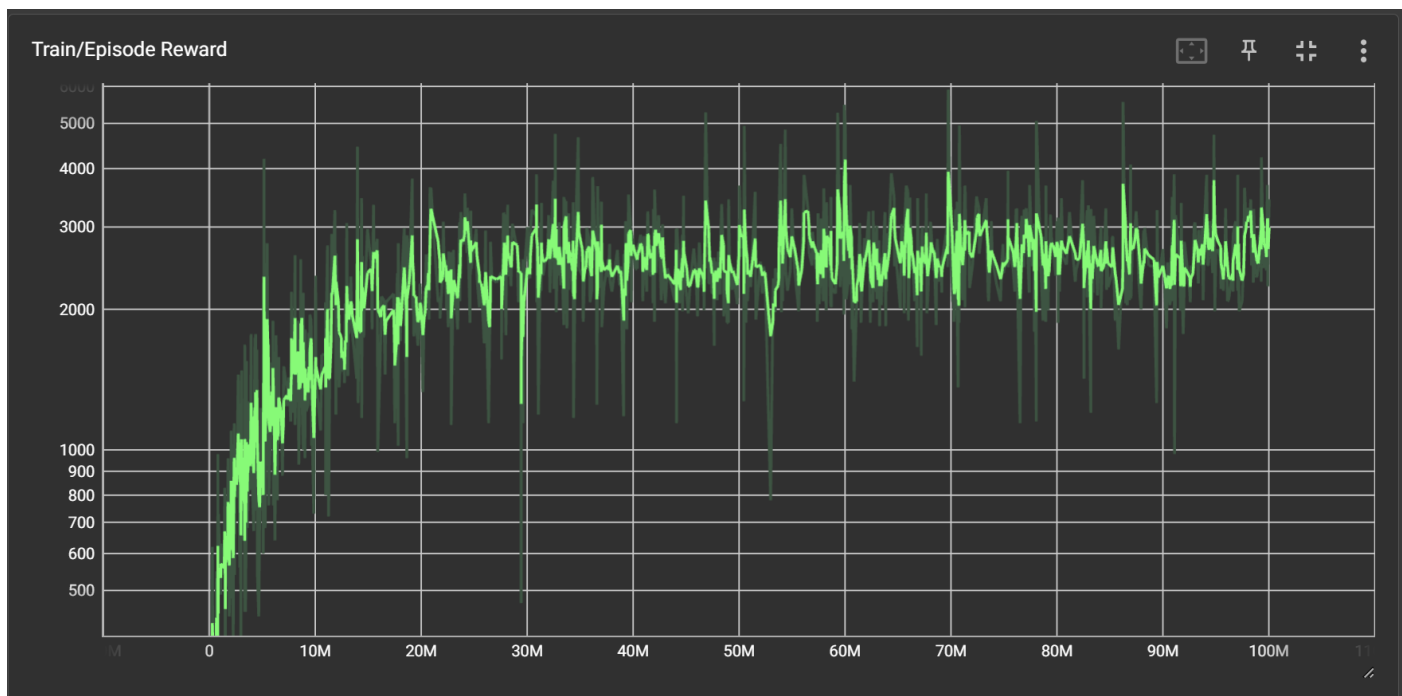
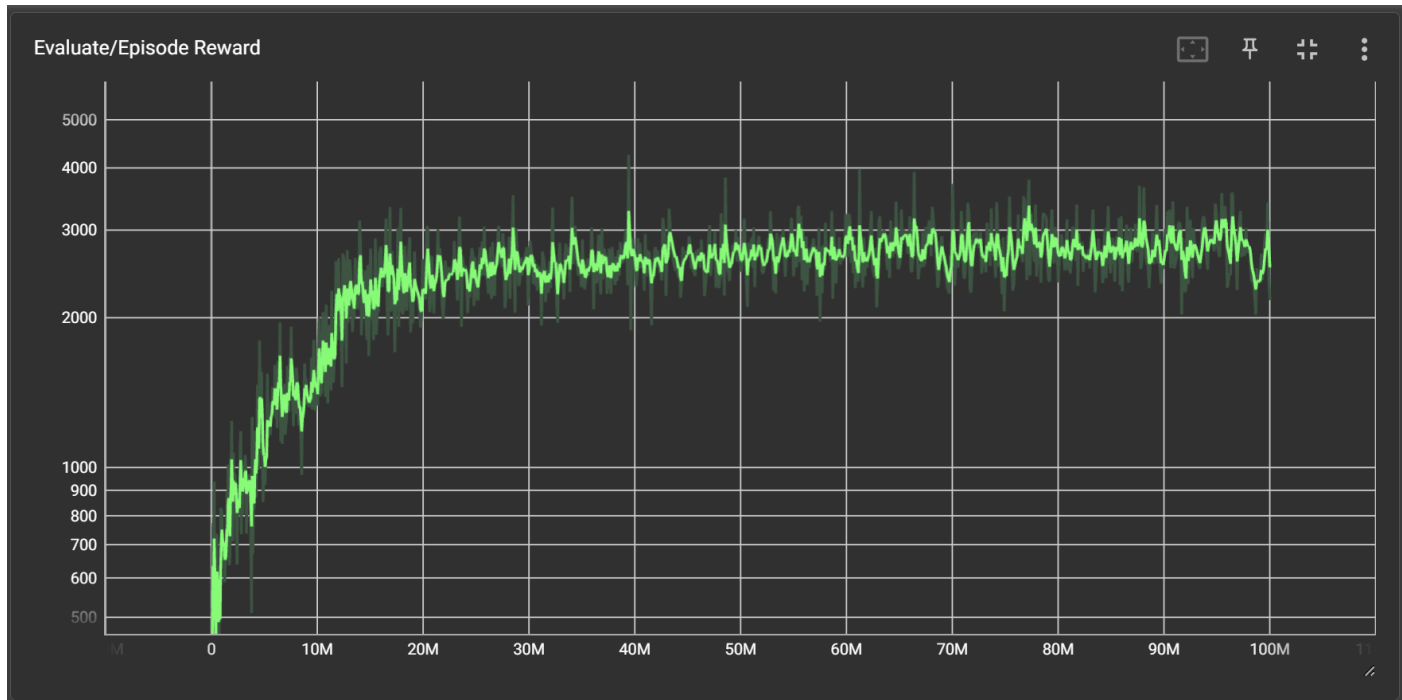
Selected topics of reinforcement learning

Lab2 – DQN

111701026 張肇政

1. Screenshot of Tensorboard training curve and testing results on DQN.

● Training curve

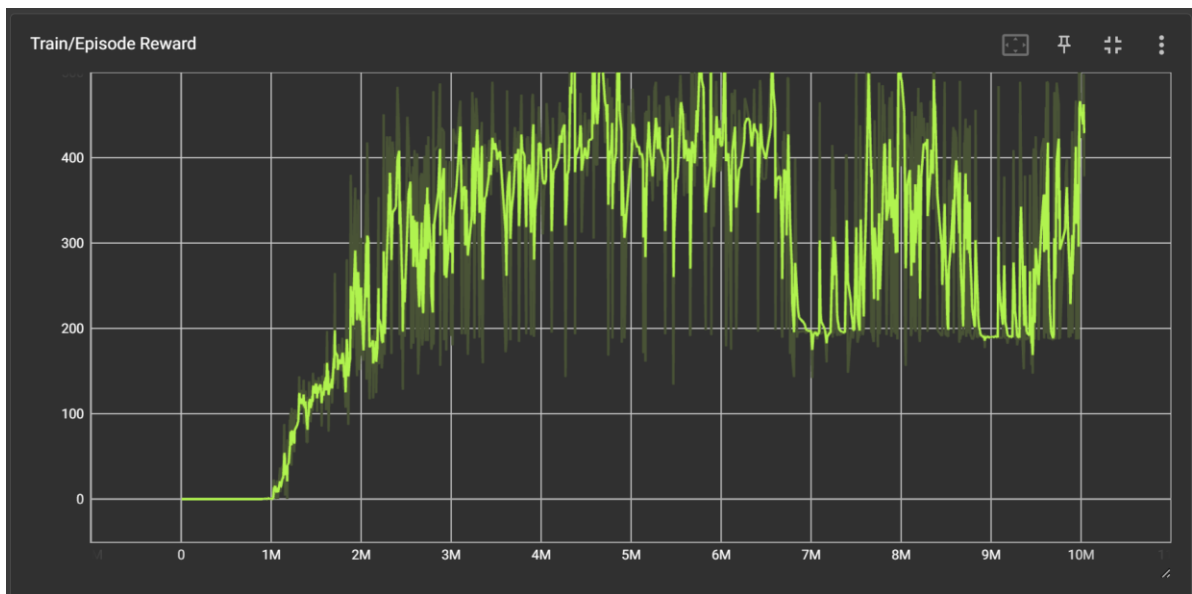
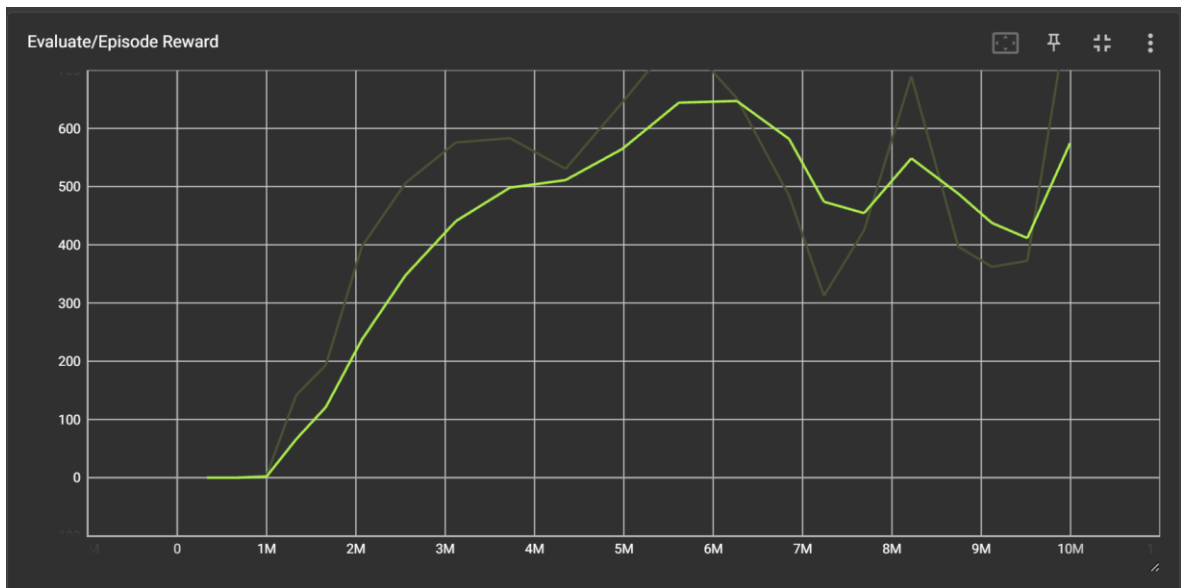


● Testing result

```
Evaluating...
episode 1 reward: 2070.0
episode 2 reward: 2240.0
episode 3 reward: 1970.0
episode 4 reward: 2090.0
episode 5 reward: 2480.0
average score: 2170.0
```

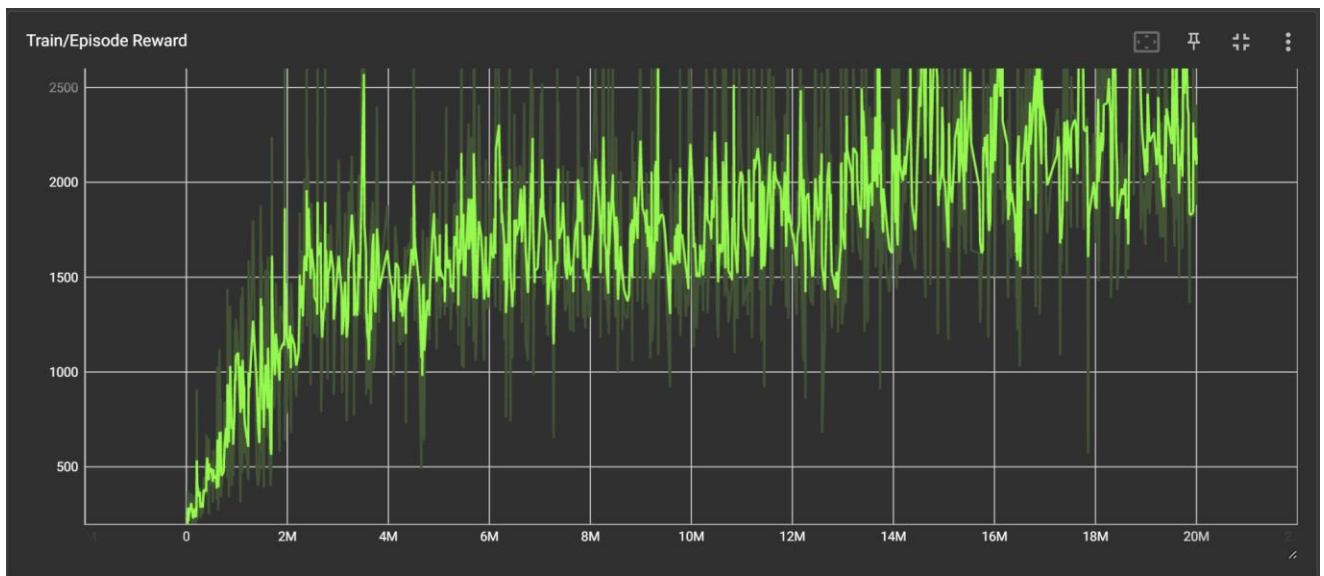
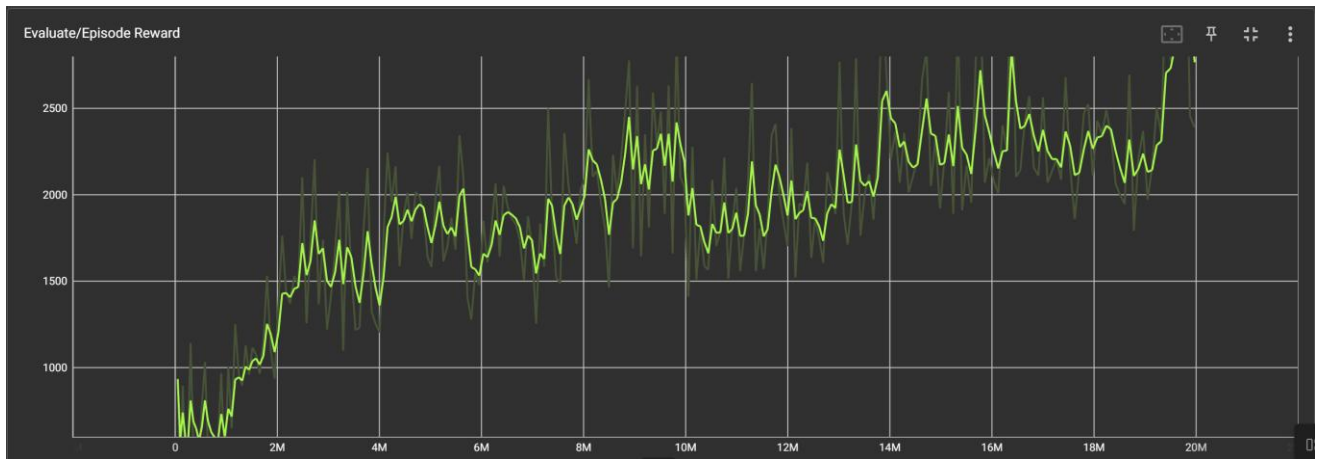
2. Others

- Screenshot of Tensorboard training curve and testing results on Enduro-v5 using DQN



```
Evaluating...
episode 1 reward: 790.0
episode 2 reward: 792.0
episode 3 reward: 1018.0
episode 4 reward: 1013.0
episode 5 reward: 485.0
average score: 819.6
```

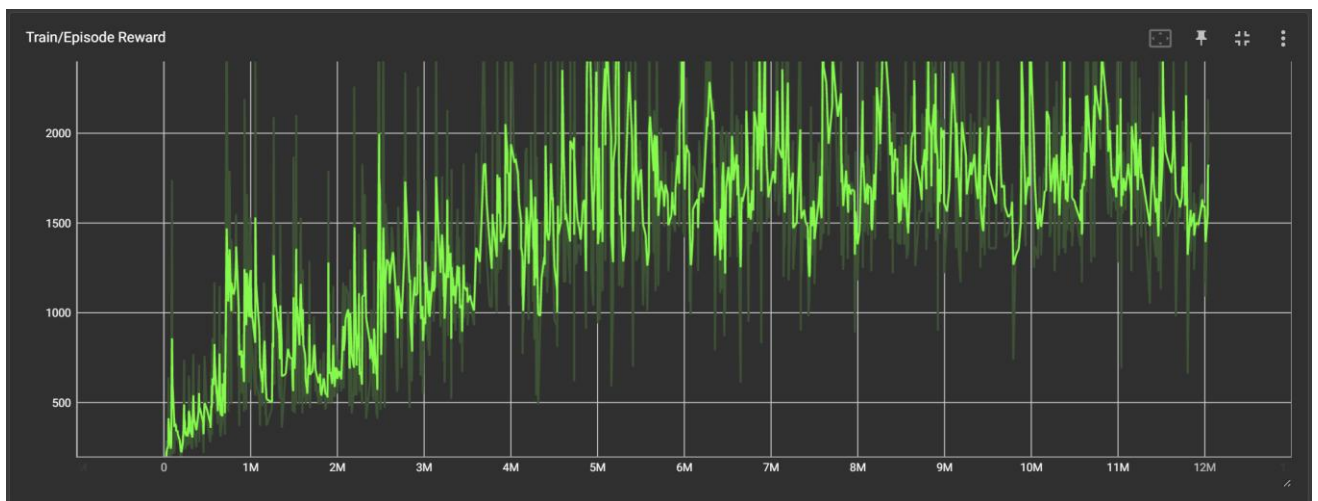
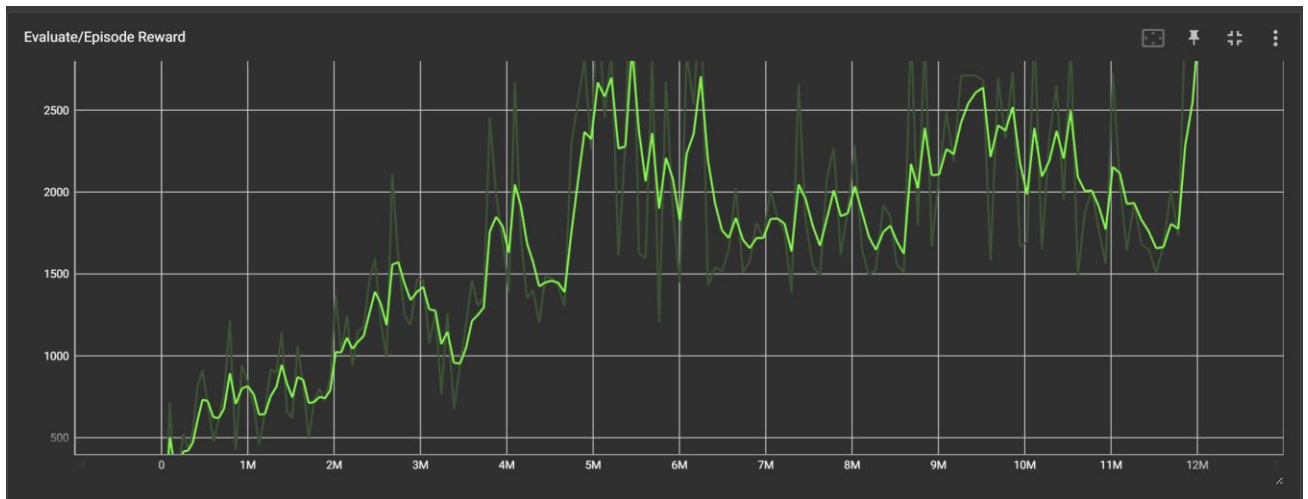
- Screenshot of Tensorboard training curve and testing results on DDQN, and discuss the difference between **DQN** and **DDQN**
 - The difference between DQN and DDQN: DQN 與 DDQN 主要在於 DDQN 能改善 DQN 對學習過程時，Q-value 高估所造成的偏差問題，DDQN 會把選擇、評估交給不同網路完成，而 DQN 是同時於同一個網路進行



```
Evaluating...
episode 1 reward: 1830.0
episode 2 reward: 2080.0
episode 3 reward: 3330.0
episode 4 reward: 2750.0
episode 5 reward: 1960.0
average score: 2390.0
```

- Screenshot of Tensorboard training curve and testing results on Dueling DQN, and discuss the difference between **DQN** and **Dueling DQN**

- The difference between DQN and Dueling DQN: Dueling DQN 的學習效率較高。DQN 的運作過程為輸入狀態 -> 卷積層 -> 全連接層 -> $[Q(s,a_1), Q(s,a_2), Q(s,a_3), \dots]$ ，而 Dueling DQN 會分成兩個，分別是 **value stream & advantage stream**，value stream 評估現在這個狀態的好壞，advantage stream 評估當前各個動作比平均好多少。



```
Evaluating...
episode 1 reward: 3450.0
episode 2 reward: 5640.0
episode 3 reward: 3420.0
episode 4 reward: 2890.0
episode 5 reward: 4200.0
average score: 3920.0
```

- Screenshot of Tensorboard training curve and testing results on DQN with parallelized rollout, and discuss the difference between **DQN** and **DQN with parallelized rollout**
 - The difference between DQN and DQN with parallelized rollout: 新增一個 `num_workers` 在 `main.py` 的 `config` 中，增加 `workers` 數量，改造成平行化 Actor-Learner 架構(沒有實作)
- Comparison with DQN、DDQN、Dueling DQN

DQN: 100M

DDQN: 20M

Dueling DQN: 12M

